

Station availability, 2020–2023

GIRA Network

Where the bike-share fails, and what EMEL should fix first

Data Analyst Project

TEAM

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Prepared for EMEL · Operations & Rebalancing



BACKGROUND

From a 100-bike pilot to a city-wide network



2017: the pilot

EMEL launched GIRA in Parque das Nações: 100 bikes, 10 stations, 1,600 volunteer testers before going public.



Built for the hills

Electric-assist bikes were part of the design from the start. Lisbon's topography made them essential, and it still shapes where stations fail.



Citywide growth

The docked network expanded from the riverside outward, station by station, across the years this project studies.



A proven habit

GIRA passed 10 million trips by 2023. Every empty dock in this data has a real rider behind it.

GIRA TODAY



147

stations by 2023 Q1, up from 85

10M+

trips since the 2017 launch

1,700+

bikes, regular and electric-assist

THE BUSINESS QUESTION

Riders keep finding stations empty or full

Riders arrive to find no bike, or no free dock, and EMEL cannot tell which stations at which hours need bikes moved first.

The operational problem, in one sentence

MAIN QUESTION

Where and when does the GIRA network fail its users, and how should EMEL prioritize rebalancing and expansion?



SQ1 • WHEN

Failure rates by hour, weekday vs weekend, and month.



SQ2 • WHERE

Which stations fail, and does Lisbon's geography explain it?



SQ3 • WHAT

Rebalancing priorities, van loops, and whether expansion aimed right.



THE DATA

Four years of station snapshots, two schemas

Every row is one station at one moment: bikes available and free docks. It is not trip data. So every question here is about availability and occupancy, never about journeys.

8 SOURCE FILES

- gira_2020_q1...q4.xlsx, one file per quarter
- gira_2021.csv, the full year and the bridge schema
- gira_2022_s1 / s2.csv, one per semester
- gira_2023_q1.xlsx, Q1 only

THE SCHEMA CHANGE

2020 tracks numdocasvacias (empty docks) and has no status column. 2022–23 tracks numdocas (total docks) plus estado. 2021 carries both, and it is the file that bridges the two eras.

8

raw source files

7.49M

raw snapshot rows

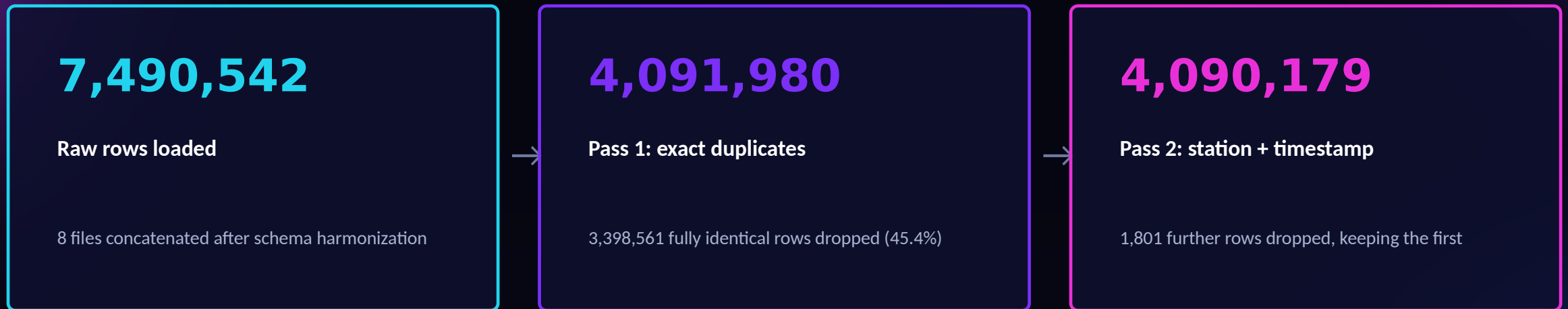
149

distinct stations

CLEANING • PART 2

7.49M raw rows → 4.09M trustworthy rows

Nearly half the rows were removed, but almost none of it was a data-quality failure. Fixed-interval logging rewrites the same station state over and over; removing those duplicates is structural, not corrective.



Final: 4,090,179 rows retained • 45.4% removed as duplicates, 0% as data loss



CLEANING • PART 2

Bridging two schemas without losing information



Validating the bridge

The 2021 file carries both dock columns, so bikes + empty docks = total docks could be tested directly. It held on 100% of 2021 rows, which is what licenses every harmonization formula built on it.



Filling the gaps both ways

2020: numdocas derived as bikes + empty docks, with an empty estado added. 2022–23: numdocasvacias derived as total docks – bikes. Every file ends with identical columns.



Two clocks, one timezone

2020 is Lisbon local time; 2021+ is UTC. Both converted to Europe/Lisbon. About 146 rows land on the DST clock-change gap and become NaT by design, not by error.



Fixing names, never overwriting

Names trimmed and a corrupted 2020 label repaired; the numeric prefix becomes a clean station_id, the reliable key for joining every year together.

CLEANING • PART 2

Flag it, don't erase it, then compute the rates

Every fix lives in a new column or boolean flag; the raw values stay available for validation.

`numdocasvacias_fix`

116 negative values clipped to 0

`is_impossible`

20,976 rows (0.51%): bikes > docks, or zero capacity

`is_repair`

160,208 rows (3.92%) under maintenance, excluded from rates

occupancy_rate is computed only after impossible rows are flagged. Set them to pd.NA first, or a divide-by-zero produces inf values that silently poison every downstream average.

4,090,179

clean rows saved

5.04%

snapshots empty

1.96%

snapshots full

85.3 MB

parquet + BOM CSV



PART 3 • ANALYSIS

Stations nearly doubled, capacity grew faster

GIRA network growth, 2020-2023

TAKEAWAYS

**85 → 102 → 146 → 147**

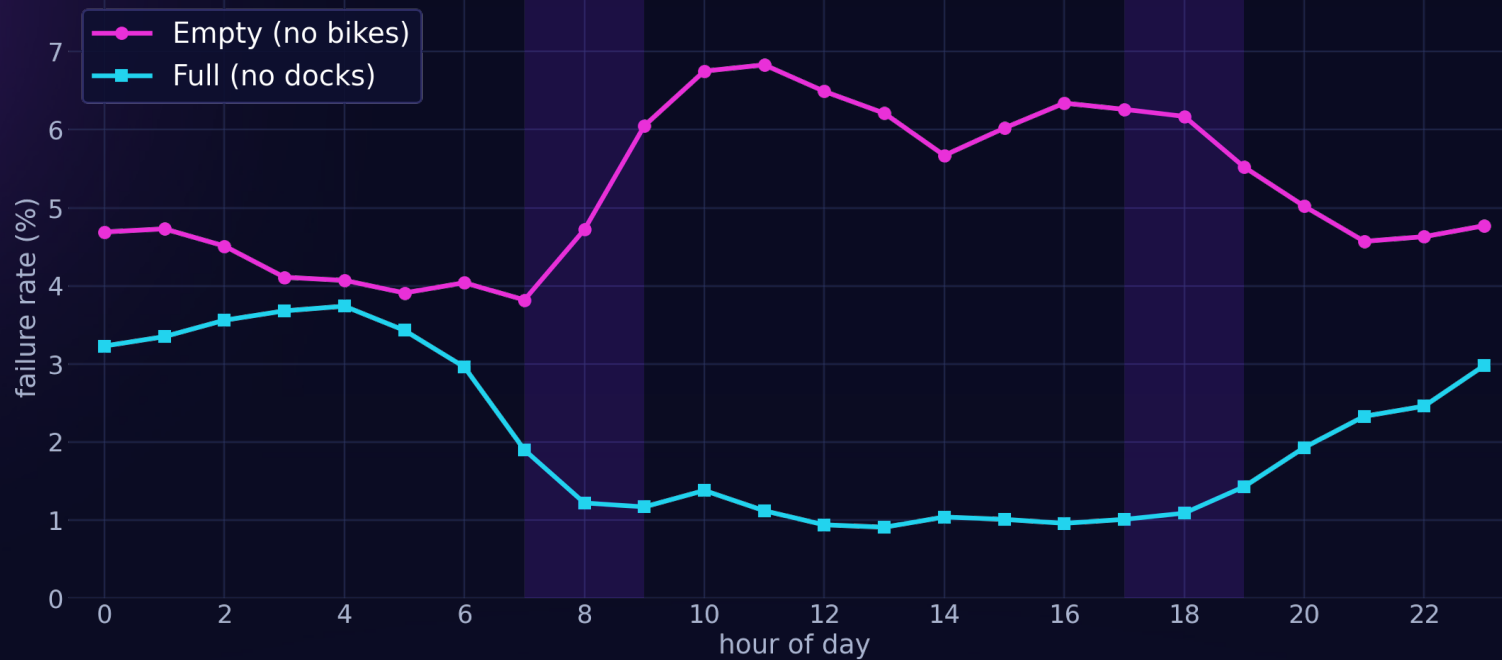
active stations, 2020 → 2023 Q1

- 2021 → 2022 capacity grew faster than station count
- newer stations were built larger, not just more numerous
- 2023 is Q1 only, so always compare like-for-like quarters

SQ1 • WHEN DOES IT FAIL?

Empty peaks mid-morning; full peaks overnight

When does GIRA fail? Failure rate by hour of day



TAKEAWAYS

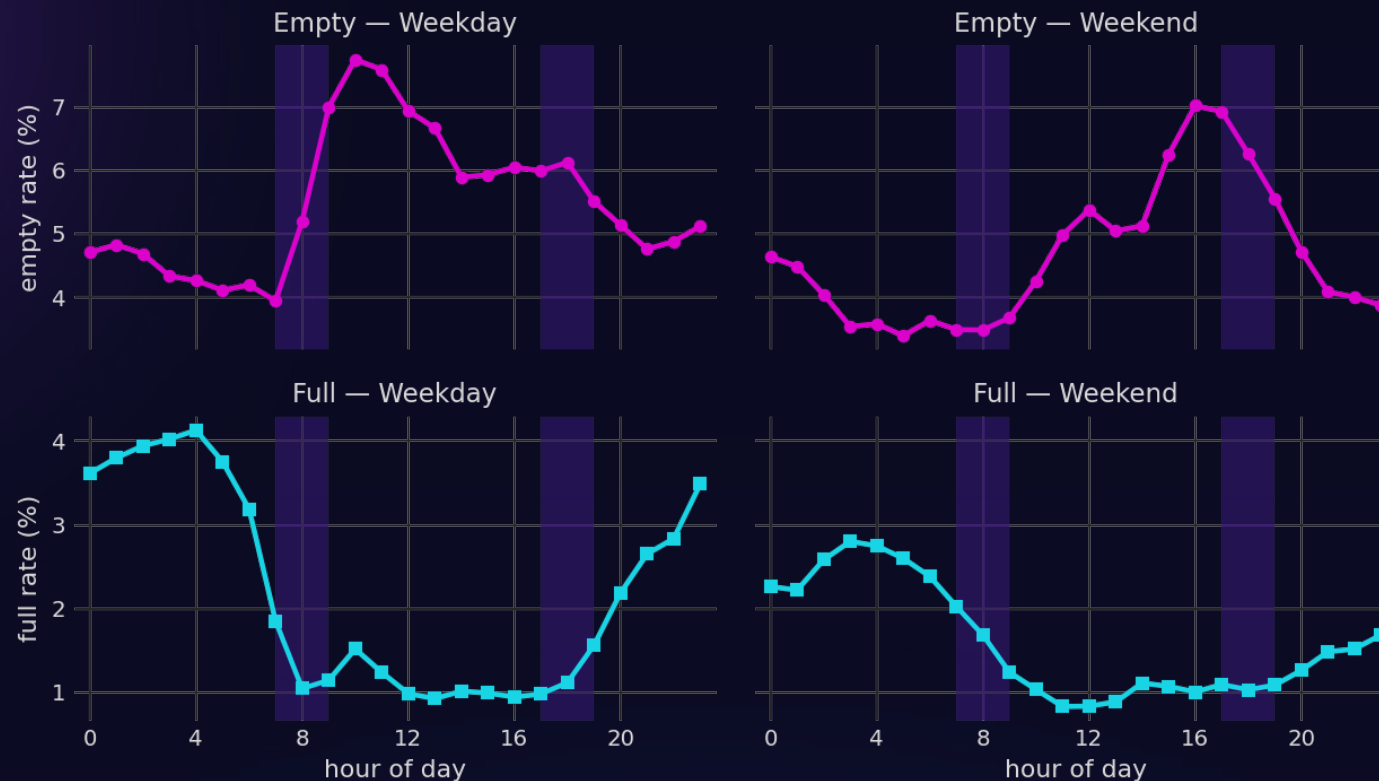
- Empty bottoms at 3.8% at 07:00, then a 6.2–6.8% plateau from 10:00 to 18:00
- Full is the mirror image: ~3.7% before dawn, under 1% by midday
- The two peak at opposite hours: bikes drain out by day and dock back up overnight

Repair and impossible rows excluded; local time throughout

SQ1 • WHEN DOES IT FAIL?

A commuter fingerprint: weekday 10:00 vs weekend 16:00

Failure patterns: weekday vs weekend



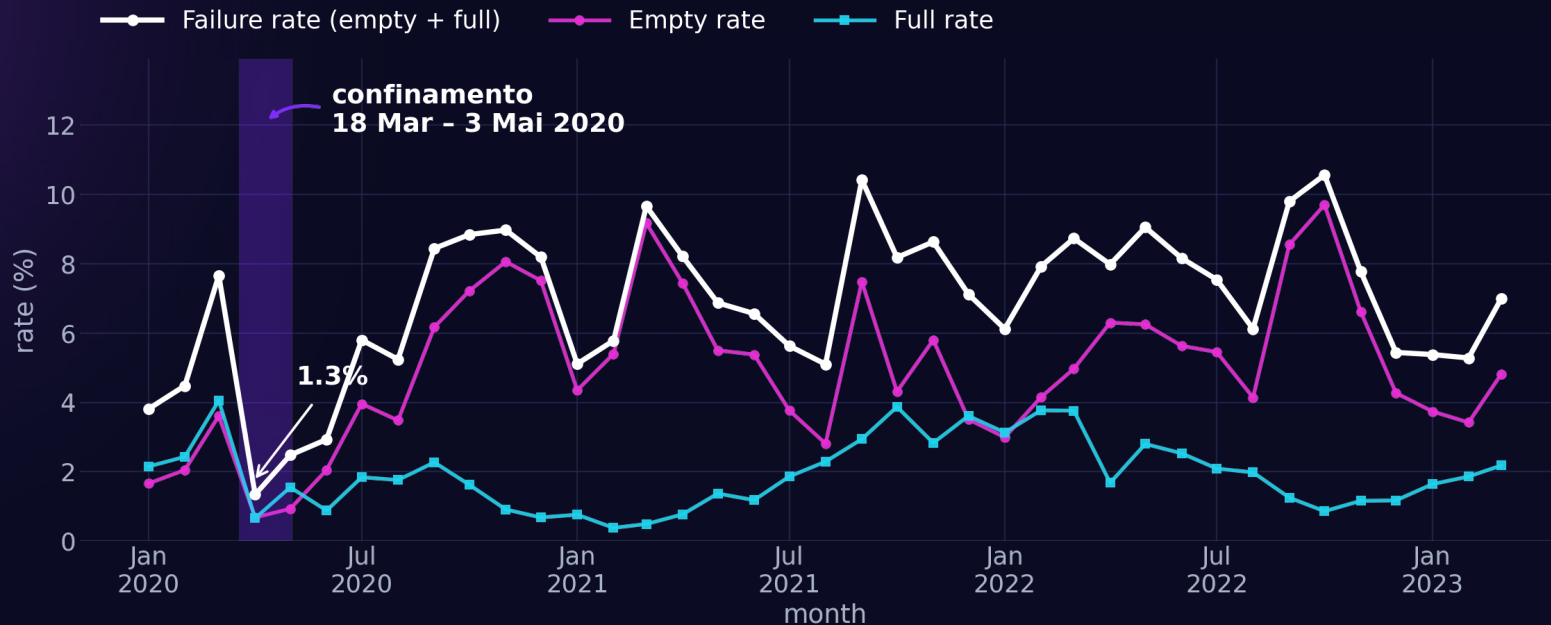
TAKEAWAYS

- Weekday empty peak: 10:00, at 7.7%
- Weekend empty peak: 16:00, at 7.0%, a full six hours later
- Weekdays swing harder (3.9–7.7%) than weekends (3.4–7.0%)
- That is the commuter mechanism: sharp weekday morning peak, gentler and later leisure pattern at weekends

SQ1 · WHEN DOES IT FAIL?

Lockdown “fixed” availability by removing riders

Monthly failure rate, 2020-2023, with the lockdown window marked



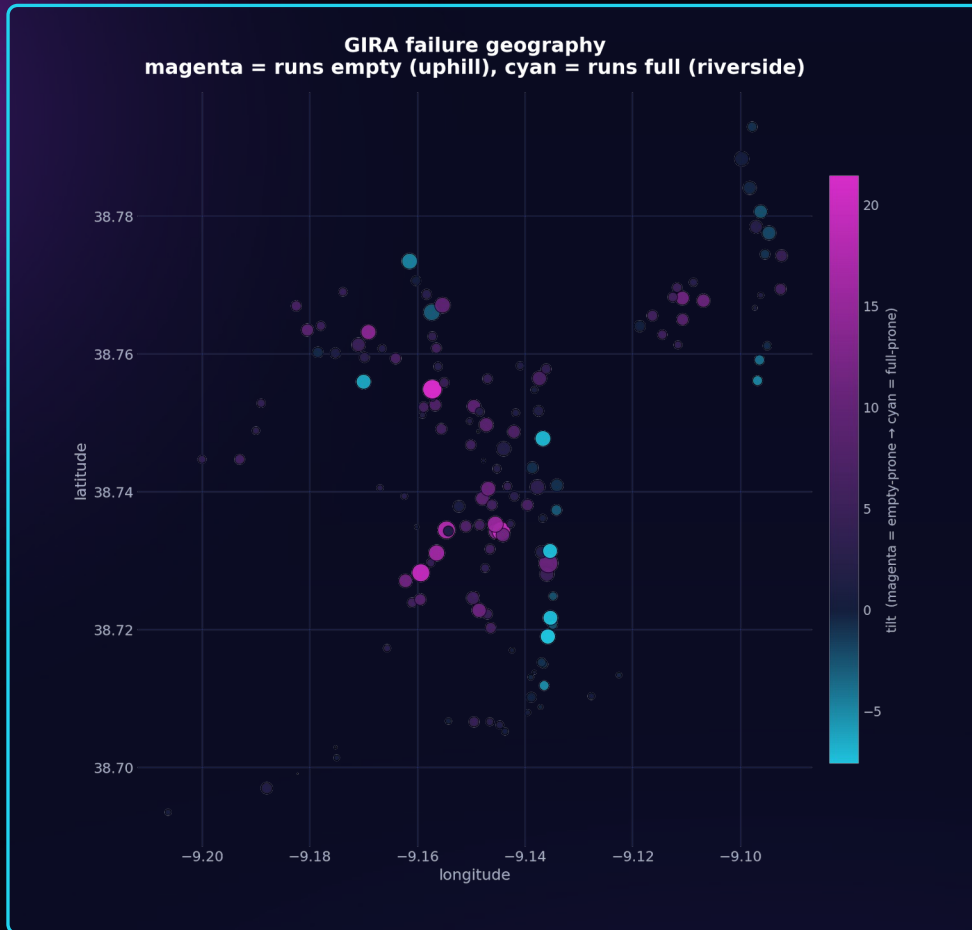
TAKEAWAYS



- April 2020 fell to 1.3%, the lowest month in the whole series
- Nobody was riding, so bikes stayed docked. Availability looked perfect because demand had vanished
- The effect lasted about four months. By September 2020 failures hit 8.4%, above the pre-lockdown 3.8–4.5%
- So, failure rate tracks demand, not network health: the rise through 2021–22 is partly a busier network

SQ2 • WHERE DOES IT FAIL, AND WHY?

Lisbon's hills write the failure map



TAKEAWAYS

0.39

tilt vs real elevation (latitude proxy managed only
0.06)

81m vs 30m

median elevation: empty-prone vs full-prone
stations



- Real elevation, joined per station from an open elevation API, is a far stronger signal than latitude. This is about altitude, not north/south
- Empty-prone stations sit uphill (magenta); full-prone sit by the river and downtown (cyan). Riders coast down and avoid the ride back up
- Worst empty: 479 – Rua Prof. Oliveira Marques, empty 21.9% of the time (~84m)
- Worst full: 452 – Rua Teixeira de Pascoais / Dr. Gama Barros, full 10.8%

Explore the failure map, station by station

The same geography finding, built as a Leaflet map you can pan, zoom and click.



141 stations

Every station with enough data, plotted on the real city map.



Colour = tilt

Red runs empty, blue runs full, on a -15 to +15 scale.



Size = severity

Bigger circle, higher combined empty + full rate.

Click any station and the popup gives its elevation, empty rate, full rate and dominant failure type.



[Open the interactive map](#)

SQ3 • WHAT SHOULD EMEL DO FIRST?

The priority list: where and when to deliver bikes

Rebalancing priority — where & when to DELIVER bikes



TOP PRIORITY

519

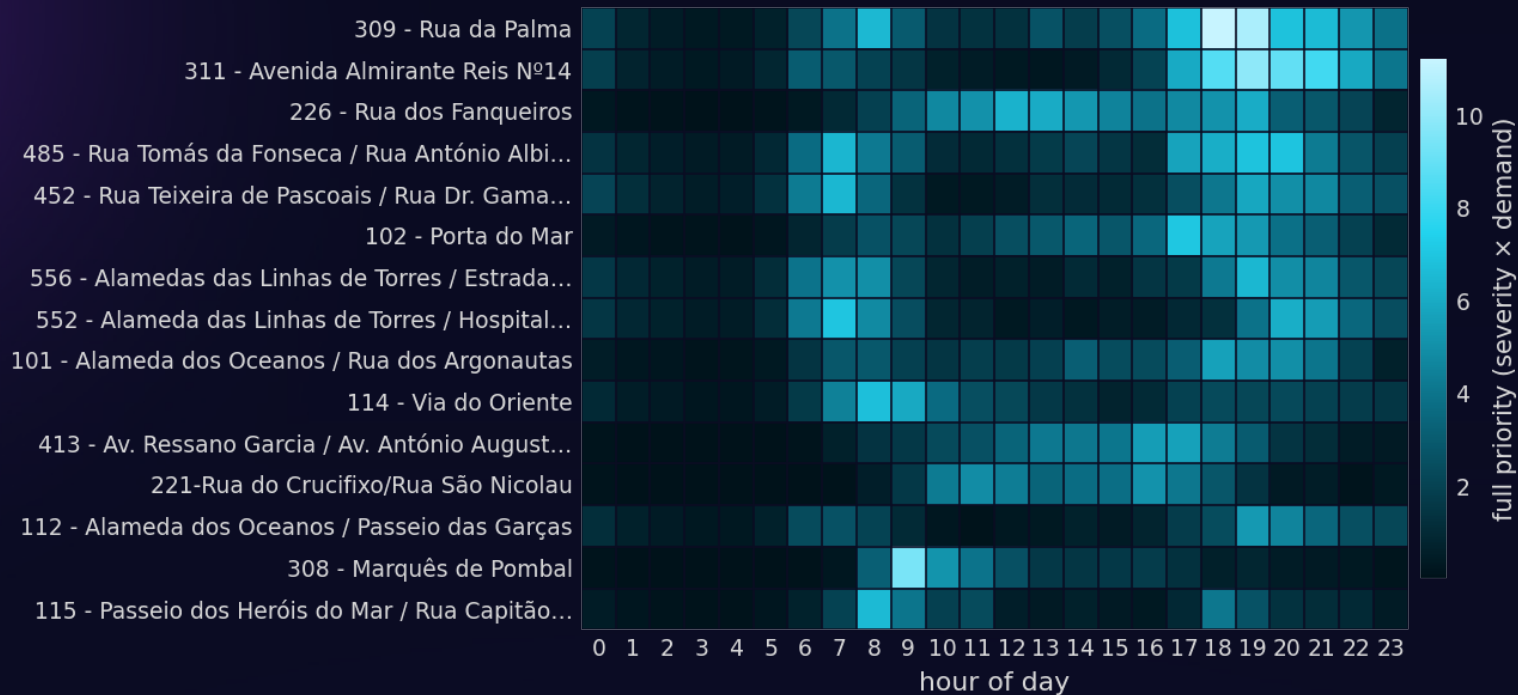
Rua Prof. Pulido Valente, 09:00,
33.3% empty

*Priority = failure severity × a
commute-shaped demand curve.
The curve is fixed, not data-derived.
Deriving demand from the same
data would be circular with the
failure rate itself.*

SQ3 • WHAT SHOULD EMEL DO FIRST?

...and where to collect, paired into short van loops

Rebalancing priority — where & when to REMOVE bikes



SAMPLE VAN LOOP



COLLECT

309 - Rua da Palma (18:00, 31 m)

↑ van climbs +48 m ↑

DELIVER

407 - Pça. Duque de Saldanha (18:00, 79 m)

The shortest hop:

COLLECT 552 - Alameda das Linhas de Torres →
DELIVER 519 - Pulido Valente

only 1.07 km, +27 m of climb

Each loop moves bikes from a low station that fills up to a higher one that runs dry. The van makes the climb riders won't.

CONCLUSIONS & RECOMMENDATIONS



What EMEL should do with this

1**Failures are concentrated**

A handful of stations, at specific hours, drive most empty and full events, not a uniform citywide shortage.

2**The failures follow commuters**

Empty peaks weekday 10:00 and weekend 16:00; full peaks overnight. A clear supply/demand rhythm.

3**Geography is the best predictor**

Elevation beats latitude by far (0.39 vs 0.06): uphill runs empty, riverside runs full.

4**Many fixes are cheap**

Several worst empty/full pairs are a short van climb apart, so loops beat fleet expansion.

5**Growth outpaced the fix**

Stations went 85→147 and capacity grew faster still, yet 479, 407, 452 and 309 remain problem stations.

6**The priority list is ready**

The demand-weighted station × hour ranking gives the rebalancing team a daily to-do list.

Thank You



*“The network doesn't fail on average. It fails at specific stations,
at specific hours, downhill by the river, and now we know which ones.”*

Python • pandas • cleaning and analysis

Power BI • live dashboard

4,090,179 clean rows • 149 stations